THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 14: **Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 36	Surface area $\propto r^2$, so $6^2 = 36$.
2	**(b)** 125	Volume $\propto a^3$, so $5^3 = 125$.
3	**(b)** $1 : 216$	Ratio of volumes $= r_1^3 : r_2^3 = 1 : 216$.
4	**(b)** 924 cm^2	$\pi r(l + r)$ with the given values.
5	**(d)** 17 cm	$l = \sqrt{8^2 + 15^2}$.
6	**(d)** 693000	$V = 693$ m^3.
7	**(d)** 72	$\frac{30}{5} \times \frac{15}{5} \times \frac{20}{5}$.
8	**(b)** 1152 cm^3	$\frac{1}{3} \times$ base area $\times$ height.
9	**(d)** 16	Surface area $\propto r^2$, so $4^2 = 16$.
10	**(b)** 216	Volume $\propto a^3$, so $6^3 = 216$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\pi r^2 h = 2079$ gives $r^2 = \frac{2079 \times 7}{22 \times 6} = 110.25$; $r = 10.5$ cm (1); CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 10.5 \times 6 = 396$ cm^2 (1).	1+1
12	$6a^2 = 294$ gives $a^2 = 49$, so $a = 7$ cm; Volume $= 7^3 = 343$ cm^3 (1); Face diagonal $= \sqrt{7^2 + 7^2} = 7\sqrt{2}$ cm (1).	1+1
13	Volume of cuboid $= 24 \times 40 \times 16 = 15360$ cm^3; Volume of one cube $= 8^3 = 512$ cm^3 (1); Number of cubes $= \frac{15360}{512} = 30$; Leftover volume $= 0$ (the cuboid divides exactly) (1).	1+1
14	$\pi r^2 h = 14784$ gives $r^2 = \frac{14784 \times 7}{22 \times 24} = 196$; $r = 14$ cm (1); CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 14 \times 24 = 2112$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$l = \sqrt{r^2 + h^2} = 17.5$ cm (1); CSA $= \pi rl = 577.5$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 1617$ cm^3 (1).	1+1+1
16	Old TSA $= 6 \times 9^2 = 486$, new TSA $= 6 \times 27^2 = 4374$ cm^2 (1); Old volume $= 729$, new volume $= 19683$ cm^3 (1); Surface area becomes $3^2 = 9$ times, volume $3^3 = 27$ times (1).	1+1+1
17	CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 21^2 = 2772$ cm^2 (1); TSA $= 3\pi r^2 = 4158$ cm^2 (2).	1+2
	OR: $l = \sqrt{r^2 + h^2} = 59.5$ cm (1); CSA $= \pi rl = 5236$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 43120$ cm^3 (1).	1+1+1
18	Old TSA $= 6 \times 4^2 = 96$, new TSA $= 6 \times 8^2 = 384$ cm^2 (1); Old volume $= 64$, new volume $= 512$ cm^3 (1); Surface area becomes $2^2 = 4$ times, volume $2^3 = 8$ times (1).	1+1+1
	OR: CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 35^2 = 7700$ cm^2 (1); TSA $= 3\pi r^2 = 11550$ cm^2 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Volume of cylinder $= \pi r^2 \times 24$ (1); Volume of one cone $= \frac{1}{3}\pi r^2 \times 1$ (2); $n = \frac{\pi r^2 \times 24}{\frac{1}{3}\pi r^2 \times 1} = \frac{3 \times 24}{1} = 72$ (2).	1+2+2
	OR: Volume of cuboid $= 70000$ cm^3; volume of a cube $= 125$ (1); Number of cubes $= 560$ (1); TSA of one cube $= 6 \times 5^2 = 150$ (1); Total surface area $= 560 \times 150 = 84000$ cm^2 (2).	1+1+1+2
20	Volume of cylinder $= \pi r^2 \times 24$ (1); Volume of one cone $= \frac{1}{3}\pi r^2 \times 1$ (2); $n = \frac{\pi r^2 \times 24}{\frac{1}{3}\pi r^2 \times 1} = \frac{3 \times 24}{1} = 72$ (2).	1+2+2
	OR: Volume of cuboid $= 11250$ cm^3; volume of a cube $= 125$ (1); Number of cubes $= 90$ (1); TSA of one cube $= 6 \times 5^2 = 150$ (1); Total surface area $= 90 \times 150 = 13500$ cm^2 (2).	1+1+1+2

Mistakes to avoid:

- Forgetting the factor $\frac{1}{3}$ in the volume of a cone or pyramid.
- Using the vertical height h instead of the slant height l in πrl (CSA of a cone).

- Writing the TSA of a cylinder as $2\pi rh + \pi r^2$; there are two circular ends, so it is $2\pi rh + 2\pi r^2$.
- Using the diameter as the radius, or leaving r in the formula when the problem gives d.
- Mixing units: converting cm^3 to litres needs $\div 1000$, and m^3 to litres needs $\times 1000$.
- For a hemisphere, using the sphere formulas without halving the volume, or taking TSA as $2\pi r^2$ and forgetting the flat circle ($3\pi r^2$ in all).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 14: **Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $1 : 125$	Ratio of volumes $= r_1^3 : r_2^3 = 1 : 125$.
2	**(a)** 704 cm^2	$\pi r(l + r)$ with the given values.
3	**(b)** 29 cm	$l = \sqrt{20^2 + 21^2}$.
4	**(b)** 600000	$V = 600$ m^3.
5	**(c)** 60	$\frac{16}{4} \times \frac{20}{4} \times \frac{12}{4}$.
6	**(c)** 768 cm^3	$\frac{1}{3} \times$ base area $\times$ height.
7	**(b)** 4	Surface area $\propto r^2$, so $2^2 = 4$.
8	**(c)** 27	Volume $\propto a^3$, so $3^3 = 27$.
9	**(b)** $1 : 27$	Ratio of volumes $= r_1^3 : r_2^3 = 1 : 27$.
10	**(b)** 8316 cm^2	$\pi r(l + r)$ with the given values.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$6a^2 = 1176$ gives $a^2 = 196$, so $a = 14$ cm; Volume $= 14^3 = 2744$ cm^3 (1); Face diagonal $= \sqrt{14^2 + 14^2} = 14\sqrt{2}$ cm (1).	1+1
12	Volume of cuboid $= 20 \times 15 \times 10 = 3000$ cm^3; Volume of one cube $= 5^3 = 125$ cm^3 (1); Number of cubes $= \frac{3000}{125} = 24$; Leftover volume $= 0$ (the cuboid divides exactly) (1).	1+1
13	$\pi r^2 h = 11088$ gives $r^2 = \frac{11088 \times 7}{22 \times 8} = 441$; $r = 21$ cm (1); CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 21 \times 8 = 1056$ cm^2 (1).	1+1
14	$6a^2 = 96$ gives $a^2 = 16$, so $a = 4$ cm; Volume $= 4^3 = 64$ cm^3 (1); Face diagonal $= \sqrt{4^2 + 4^2} = 4\sqrt{2}$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$l = \sqrt{r^2 + h^2} = 70$ cm (1); CSA $= \pi rl = 9240$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 103488$ cm^3 (1).	1+1+1
16	Old TSA $= 6 \times 6^2 = 216$, new TSA $= 6 \times 12^2 = 864$ cm^2 (1); Old volume $= 216$, new volume $= 1728$ cm^3 (1); Surface area becomes $2^2 = 4$ times, volume $2^3 = 8$ times (1).	1+1+1
17	CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 14^2 = 1232$ cm^2 (1); TSA $= 3\pi r^2 = 1848$ cm^2 (2).	1+2
	OR: $l = \sqrt{r^2 + h^2} = 25$ cm (1); CSA $= \pi rl = 550$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 1232$ cm^3 (1).	1+1+1
18	Old TSA $= 6 \times 8^2 = 384$, new TSA $= 6 \times 24^2 = 3456$ cm^2 (1); Old volume $= 512$, new volume $= 13824$ cm^3 (1); Surface area becomes $3^2 = 9$ times, volume $3^3 = 27$ times (1).	1+1+1
	OR: CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 3.5^2 = 77$ cm^2 (1); TSA $= 3\pi r^2 = 115.5$ cm^2 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Volume of cylinder $= \pi r^2 \times 15$ (1); Volume of one cone $= \frac{1}{3}\pi r^2 \times 9$ (2); $n = \frac{\pi r^2 \times 15}{\frac{1}{3}\pi r^2 \times 9} = \frac{3 \times 15}{9} = 5$ (2).	1+2+2
	OR: Volume of cuboid $= 109375$ cm^3; volume of a cube $= 125$ (1); Number of cubes $= 875$ (1); TSA of one cube $= 6 \times 5^2 = 150$ (1); Total surface area $= 875 \times 150 = 131250$ cm^2 (2).	1+1+1+2
20	Volume of cylinder $= \pi r^2 \times 24$ (1); Volume of one cone $= \frac{1}{3}\pi r^2 \times 2$ (2); $n = \frac{\pi r^2 \times 24}{\frac{1}{3}\pi r^2 \times 2} = \frac{3 \times 24}{2} = 36$ (2).	1+2+2
	OR: Volume of cuboid $= 16384$ cm^3; volume of a cube $= 64$ (1); Number of cubes $= 256$ (1); TSA of one cube $= 6 \times 4^2 = 96$ (1); Total surface area $= 256 \times 96 = 24576$ cm^2 (2).	1+1+1+2

Mistakes to avoid:

- Forgetting the factor $\frac{1}{3}$ in the volume of a cone or pyramid.
- Using the vertical height h instead of the slant height l in πrl (CSA of a cone).

- Writing the TSA of a cylinder as $2\pi rh + \pi r^2$; there are two circular ends, so it is $2\pi rh + 2\pi r^2$.
- Using the diameter as the radius, or leaving r in the formula when the problem gives d.
- Mixing units: converting cm^3 to litres needs $\div 1000$, and m^3 to litres needs $\times 1000$.
- For a hemisphere, using the sphere formulas without halving the volume, or taking TSA as $2\pi r^2$ and forgetting the flat circle ($3\pi r^2$ in all).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 14: **Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 7 cm	$r = \sqrt{25^2 - 24^2}$.
2	**(a)** 210000	$V = 210$ m^3.
3	**(c)** 48	$\frac{10}{5} \times \frac{30}{5} \times \frac{20}{5}$.
4	**(c)** 1210 cm^3	$\frac{1}{3} \times$ base area $\times$ height.
5	**(b)** 25	Surface area $\propto r^2$, so $5^2 = 25$.
6	**(c)** 64	Volume $\propto a^3$, so $4^3 = 64$.
7	**(b)** 1 : 8	Ratio of volumes $= r_1^3 : r_2^3 = 1 : 8$.
8	**(a)** 14784 cm^2	$\pi r(l + r)$ with the given values.
9	**(b)** 75 cm	$l = \sqrt{21^2 + 72^2}$.
10	**(c)** 360000	$V = 360$ m^3.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume of cuboid $= 15 \times 9 \times 18 = 2430$ cm^3; Volume of one cube $= 3^3 = 27$ cm^3 (1); Number of cubes $= \frac{2430}{27} = 90$; Leftover volume $= 0$ (the cuboid divides exactly) (1).	1+1
12	$\pi r^2 h = 30492$ gives $r^2 = \frac{30492 \times 7}{22 \times 22} = 441$; $r = 21$ cm (1); CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 21 \times 22 = 2904$ cm^2 (1).	1+1
13	$6a^2 = 486$ gives $a^2 = 81$, so $a = 9$ cm; Volume $= 9^3 = 729$ cm^3 (1); Face diagonal $= \sqrt{9^2 + 9^2} = 9\sqrt{2}$ cm (1).	1+1
14	Volume of cuboid $= 15 \times 25 \times 20 = 7500$ cm^3; Volume of one cube $= 5^3 = 125$ cm^3 (1); Number of cubes $= \frac{7500}{125} = 60$; Leftover volume $= 0$ (the cuboid divides exactly) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$l = \sqrt{r^2 + h^2} = 50$ cm (1); CSA $= \pi rl = 2200$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 9856$ cm^3 (1).	1+1+1
16	Old TSA $= 6 \times 2^2 = 24$, new TSA $= 6 \times 10^2 = 600$ cm^2 (1); Old volume $= 8$, new volume $= 1000$ cm^3 (1); Surface area becomes $5^2 = 25$ times, volume $5^3 = 125$ times (1).	1+1+1
17	CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 28^2 = 4928$ cm^2 (1); TSA $= 3\pi r^2 = 7392$ cm^2 (2).	1+2
	OR: $l = \sqrt{r^2 + h^2} = 52.5$ cm (1); CSA $= \pi rl = 5197.5$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 43659$ cm^3 (1).	1+1+1
18	Old TSA $= 6 \times 4^2 = 96$, new TSA $= 6 \times 12^2 = 864$ cm^2 (1); Old volume $= 64$, new volume $= 1728$ cm^3 (1); Surface area becomes $3^2 = 9$ times, volume $3^3 = 27$ times (1).	1+1+1
	OR: CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 7^2 = 308$ cm^2 (1); TSA $= 3\pi r^2 = 462$ cm^2 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Volume of cylinder $= \pi r^2 \times 24$ (1); Volume of one cone $= \frac{1}{3}\pi r^2 \times 2$ (2); $n = \frac{\pi r^2 \times 24}{\frac{1}{3}\pi r^2 \times 2} = \frac{3 \times 24}{2} = 36$ (2).	1+2+2
	OR: Volume of cuboid $= 200000$ cm^3; volume of a cube $= 125$ (1); Number of cubes $= 1600$ (1); TSA of one cube $= 6 \times 5^2 = 150$ (1); Total surface area $= 1600 \times 150 = 240000$ cm^2 (2).	1+1+1+2
20	Volume of cylinder $= \pi r^2 \times 15$ (1); Volume of one cone $= \frac{1}{3}\pi r^2 \times 9$ (2); $n = \frac{\pi r^2 \times 15}{\frac{1}{3}\pi r^2 \times 9} = \frac{3 \times 15}{9} = 5$ (2).	1+2+2
	OR: Volume of cuboid $= 7168$ cm^3; volume of a cube $= 64$ (1); Number of cubes $= 112$ (1); TSA of one cube $= 6 \times 4^2 = 96$ (1); Total surface area $= 112 \times 96 = 10752$ cm^2 (2).	1+1+1+2

Mistakes to avoid:

- Forgetting the factor $\frac{1}{3}$ in the volume of a cone or pyramid.

- Using the vertical height h instead of the slant height l in $\pi r l$ (CSA of a cone).
- Writing the TSA of a cylinder as $2\pi r h + \pi r^2$; there are two circular ends, so it is $2\pi r h + 2\pi r^2$.
- Using the diameter as the radius, or leaving r in the formula when the problem gives d.
- Mixing units: converting cm^3 to litres needs $\div 1000$, and m^3 to litres needs $\times 1000$.
- For a hemisphere, using the sphere formulas without halving the volume, or taking TSA as $2\pi r^2$ and forgetting the flat circle ($3\pi r^2$ in all).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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